CLINICAL · EAR

# Otitis media: acute and with effusion

The same small space behind the drum produces two very different patients: a feverish child in pain with a bulging drum, and a quiet child with fluid who simply does not hear well. The first needs pain relief and judgement about antibiotics; the second needs patience, a hearing test, and sometimes a grommet.

**AT A GLANCE**

- Acute disease is about pain and infection; effusion is about hearing and development.
- Pneumatic otoscopy makes the diagnosis in both.
- Analgesia for all; antibiotics by age and severity, or watch at 48–72 hours.
- Watch effusion for 3 months; offer tubes for lasting loss.
- In an adult with one-sided effusion, look at the nasopharynx.
- NICE: watch non-severe AOM; prophylaxis halves attacks in recurrent AOM; grommets or hearing aids after 3 months of bilateral OME with ≥25–30 dB loss.

## 1 CHAPTER ONE Overview

Two diseases of one cleft.

### 1.1 Definitions

#### 1.1.1 Acute otitis media

Rapid onset, within 48 hours, of middle ear inflammation with an effusion: pain, fever and a red, bulging drum.

#### 1.1.2 Recurrent acute otitis media

Three or more episodes in six months, or four or more in a year, with the child well in between.

#### 1.1.3 Otitis media with effusion

Fluid in the middle ear behind an intact drum, without signs of acute infection. Also called *secretory otitis media* or *glue ear*. It is *chronic* once it has lasted three months.

#### 1.1.4 Acute necrotising otitis media

> A rare, destructive form, usually from beta-haemolytic streptococcus after measles or scarlet fever, which rapidly destroys the drum, annulus and ossicles.

### 1.2 Epidemiology

Acute otitis media is one of the commonest reasons a young child sees a doctor, peaking between six months and two years. Effusion follows most acute attacks and is the commonest cause of hearing loss in childhood, with peaks around two and again around five years.

### 1.3 Why it matters

Acute disease is about pain and the rare but serious complication. Effusion is about hearing, speech, behaviour and school, in a child who often does not complain at all.

## 2 CHAPTER TWO Pathogenesis

Why children, and why after a cold.

### 2.1 The Eustachian tube in a child

The child's tube is shorter, wider and more horizontal than the adult's, and its muscles open it less efficiently. Secretions from the nasopharynx reach the middle ear easily, and the middle ear drains poorly.

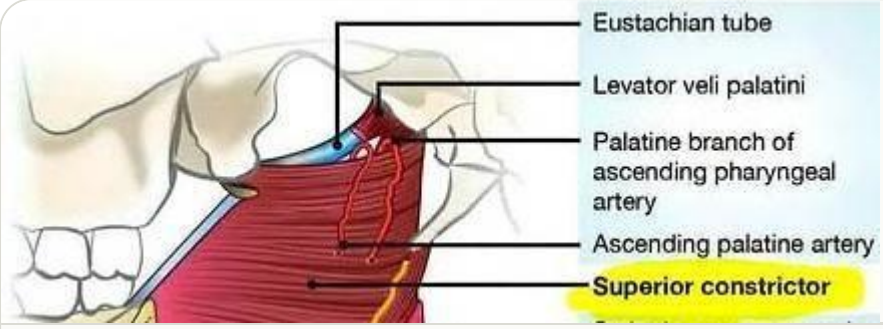



**FIGURE 1** The Eustachian tube with tensor and levator veli palatini, the muscles that open it.
Anatomy atlas · ENT Steps p. 204

## 2.2 The sequence

A viral cold inflames the tube and the adenoid. The tube blocks, gas in the middle ear is absorbed, negative pressure develops and a transudate collects. Bacteria from the nasopharynx, where the adenoid acts as a reservoir, then infect the fluid.

## 2.3 Organisms

### 2.3.1 Bacteria

*Streptococcus pneumoniae* leads, followed by non-typeable *Haemophilus influenzae* and *Moraxella catarrhalis*. About half of the *H. influenzae* and almost all *Moraxella* produce beta-lactamase.

### 2.3.2 Viruses

RSV, rhinovirus and adenovirus are found in many aspirates, alone or with bacteria.

## 2.4 Risk factors

**WHO GETS OTITIS MEDIA**

- ENVIRONMENT **Daycare, siblings** smoke exposure, dummy use, bottle feeding lying flat, no breastfeeding
- ANATOMY **Cleft palate** Down syndrome, craniofacial syndromes
- HOST **Prematurity** immunodeficiency, low socioeconomic status
- ADENOID **Bacterial reservoir** and tubal obstruction

# 3 CHAPTER THREE Acute otitis media

Pain, fever and a bulging drum.

## 3.1 Stages

Dhingra describes acute suppurative otitis media as a sequence, and the drum changes at each stage.

### 3.1.1 Tubal occlusion

The drum is retracted with a foreshortened handle of malleus; there is deafness and fullness, and little fever.

### 3.1.2 Presuppuration

Hyperaemia spreads from the handle of malleus and the periphery in a *cart-wheel* pattern. Pain is throbbing, and a child may have a high fever.

### 3.1.3 Suppuration

Pus collects under pressure: the drum is red and bulging, landmarks vanish, and a yellow spot may mark where it will rupture. Pain is severe.

### 3.1.4 Resolution

The drum ruptures, usually with a small anteroinferior perforation, discharge appears and pain and fever settle. Most perforations heal.

### 3.1.5 Complication

If infection spreads beyond the cleft: mastoiditis, facial palsy, labyrinthitis or intracranial disease.

## 3.2 Clinical picture

Otalgia, fever, irritability, ear pulling and poor feeding, often after a cold. These symptoms are non-specific on their own; the diagnosis rests on the drum.

## 3.3 Diagnosis

*Pneumatic otoscopy* is the essential examination. A bulging drum with poor mobility makes the diagnosis. A red drum in a crying child is not enough, because crying flushes the drum. Tympanometry helps when the view is poor.

**LOOK FOR A COMPLICATION FIRST**

- Facial weakness, vertigo, mastoid tenderness or swelling, neck stiffness, drowsiness.
- A child with a cochlear implant, immunodeficiency or a septic appearance is treated, not observed.

## 3.4 Management

### 3.4.1 Pain relief for everyone

Paracetamol or ibuprofen, whatever is decided about antibiotics.

### 3.4.2 Observation or antibiotics

Many children recover without antibiotics. Watchful waiting for 48 to 72 hours, with a safety net, suits children over two with mild symptoms or an uncertain diagnosis, provided the parents can return. It does not raise the rate of mastoiditis.

**WHEN TO GIVE ANTIBIOTICS**

| AGE | SEVERE | NON-SEVERE, BOTH EARS | NON-SEVERE, ONE EAR |
|---|---|---|---|
| 6–23 months | Antibiotic | Antibiotic | Antibiotic or observe |
| 2 years and over | Antibiotic | Antibiotic or observe | Antibiotic or observe |

Severe means fever over 39°C or moderate to severe pain for more than 48 hours.

### 3.4.3 Choice of antibiotic

- **First line:** high-dose amoxicillin.
- **Amoxicillin-clavulanate:** if amoxicillin was given in the last 30 days, with purulent conjunctivitis, or after recurrent failure.
- **Failure at 48 to 72 hours:** step up to amoxicillin-clavulanate, then intramuscular ceftriaxone for three days, then tympanocentesis for culture.

WHERE THE BOOKS DIFFER · DOSE AND DURATION OF AMOXICILLIN

| Dhingra | Lee |
|---|---|
| 40 mg/kg/day for at least 10 days. | 90 mg/kg/day (high dose, to overcome resistant pneumococci), with duration by age. |

Current guidelines follow the high-dose regimen; know both for examinations.

### 3.4.4 Myringotomy

Indicated for a bulging drum with severe pain, incomplete resolution with persisting hearing loss, or effusion beyond 12 weeks, and in the child who fails second-line treatment, when the fluid is sent for culture.

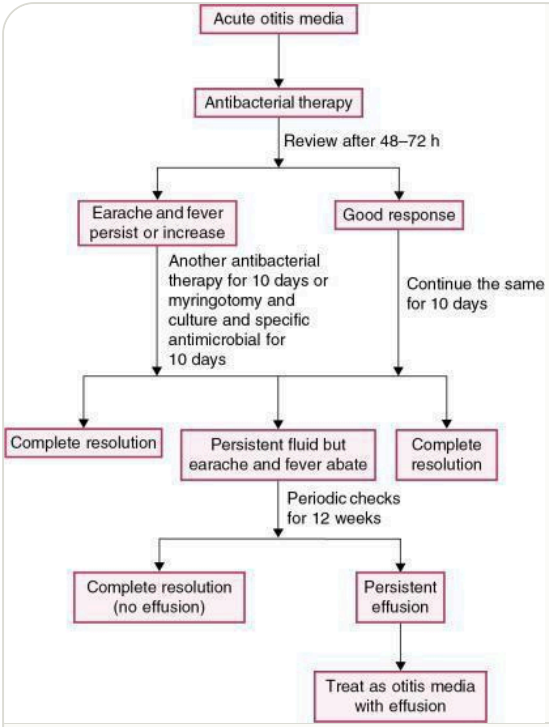



**FIGURE 2** Treatment of acute otitis media, matched to its stage.
Dhingra, fig. 10.1

## 3.5 Recurrent acute otitis media

Prophylactic antibiotics are no longer recommended; grommets reduce the number of episodes, and adenoidectomy is added in older children or for repeat surgery.

WHERE THE BOOKS DIFFER · RECURRENT OTITIS MEDIA

| Dhingra | Lee |
|---|---|

| | |
|---|---|
| Mentions amoxicillin prophylaxis, while noting its drawbacks and that many prefer grommets; grommets for 4 episodes in 6 months or 6 in a year. | Does not recommend prophylaxis; defines recurrence as 3 in 6 months or 4 in 12 months. |

**ACUTE AND RECURRENT OTITIS MEDIA: GUIDANCE AND EVIDENCE (OXFORD TEXTBOOK)**

**Antibiotics.** NICE (2018) advises analgesia and self-care for non-severe acute otitis media, including unilateral disease from 6 to 23 months, because most cases resolve and the number of children treated to prevent one case of mastoiditis is about 4,800. Immediate antibiotics are reserved for children under two with bilateral disease, those with otorrhoea, the systemically unwell, those under six months and the immunocompromised: amoxicillin first, a macrolide in penicillin allergy, and high-dose co-amoxiclav if symptoms persist. The American Academy of Pediatrics allows shared decision-making with a 48 to 72 hour wait-and-see option.

**Recurrent acute otitis media** is three or more episodes in six months or four or more in a year. Risk factors include day care, parental smoking and dummies; breastfeeding protects, and pneumococcal and Hib vaccination reduce episodes. Children who keep having infections despite vaccination may have poor antibody responses and benefit from a booster with the 23-valent vaccine. A meta-analysis of 14 trials found that long-term **antibiotic prophylaxis** (amoxicillin at half dose twice daily, or azithromycin 10 mg/kg three times a week, for two to three months) cut episodes from about three to 1.5 a year, with about five children treated to prevent one child having an attack, although the effect stops with the drug and resistance is a concern. **Grommets** reduce episodes by about one in the following six months, with little effect by a year, and allow infections to be treated with quinolone drops; they are the usual choice when effusion persists between attacks. Direct comparisons of prophylaxis and grommets are inconclusive.

**Complications** now occur in only 0.05 to 0.25% of episodes. Acute mastoiditis follows fewer than 0.4%, mostly under two years; intravenous antibiotics alone succeed in about 73%, with myringotomy, grommet, drainage of a subperiosteal abscess or cortical mastoidectomy for the rest.

CHAPTER FOUR

# 4 Otitis media with effusion

The quiet ear that cannot hear.

## 4.1 Pathogenesis

Tubal dysfunction leaves negative pressure and fluid behind; the mucosa responds with more goblet cells and mucous glands, and the fluid becomes thick and tenacious, the classic *glue*. Persistent adenoid infection, allergy and unresolved acute otitis media all contribute.

## 4.2 Clinical picture

### 4.2.1 Symptoms

Often none: a parent or teacher notices inattention, poor speech, or a child turning up the television. Older children and adults describe fullness and muffled hearing.

### 4.2.2 Signs

The drum is dull and opaque, sometimes amber, grey or bluish, with fine radial vessels rather than redness. It is often retracted, and may show an air-fluid level or bubbles. Mobility on pneumatic otoscopy is reduced.



**FIGURE 3** Otitis media with effusion: bubbles appear behind the drum on Valsalva.
Dhingra, fig. 10.2

## 4.3 Investigations

### 4.3.1 Tympanometry

A flat (type B) curve, or one shifted far to the negative side (type C).

### 4.3.2 Audiometry

Usually a 20 to 40 dB conductive loss. An audiogram is needed when the effusion has lasted three months, or sooner if there is a speech or developmental concern.

### 4.3.3 The nasopharynx in adults

**UNILATERAL EFFUSION IN AN ADULT**

Always examine the nasopharynx: nasopharyngeal carcinoma can first present this way, especially in smokers and people from endemic regions.

**WHERE THE BOOKS DIFFER · SCOPING THE NASOPHARYNX**

**Dhingra**

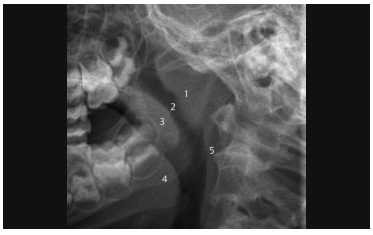
Always exclude a nasopharyngeal tumour in an adult with one-sided effusion.

**Lee**

The value of endoscopy is clearest in high-risk patients.

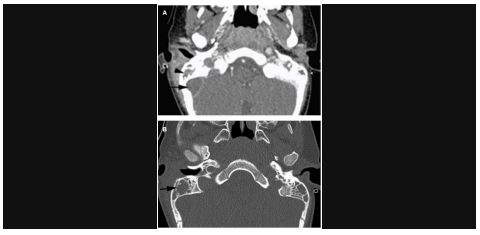
**IMAGING IN THE APP** 2 cases

Adenoid enlargement, and otomastoiditis with its complications.



**Adenoid Hypertrophy**

Infection & inflammation · 1 image



**Otomastoiditis and its complications**

Infection & inflammation · 9 images

## 4.4 Natural history

Most effusions clear on their own: about 70% within three months, and around 90%

after treated acute otitis media.

## 4.5 Management

### 4.5.1 Watchful waiting

Review every three months with a hearing test. Antibiotics, steroids, antihistamines and decongestants have no lasting benefit and are not recommended. Autoinflation with a nasal balloon helps some children.

**WHERE THE BOOKS DIFFER · MEDICAL TREATMENT OF EFFUSION**

| Dhingra | Lee |
|---|---|
| Lists decongestants, antihistamines and antibiotics among the medical options. | Guidelines find no role for them. |

Follow current guidelines in practice.

### 4.5.2 Grommets

For effusion lasting three months or more with hearing loss, or in a child at risk (speech delay, learning or sensory difficulty, cleft palate, Down syndrome). Thick glue may need two incisions, one to let air in behind the fluid.

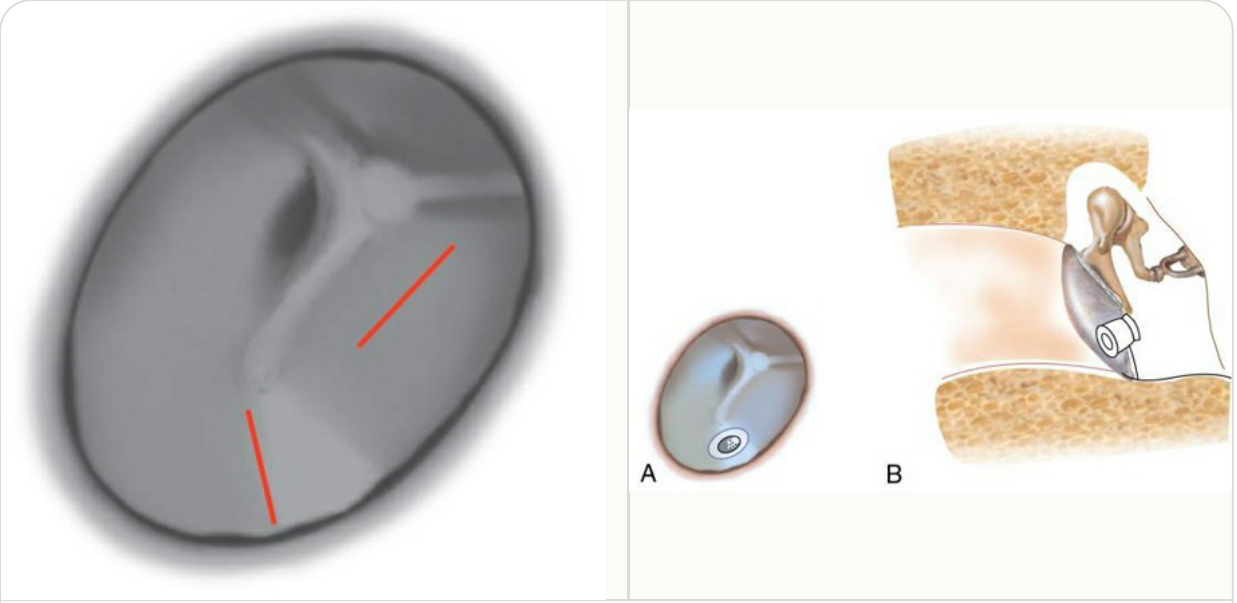



**FIGURE 4** First, two incisions in the drum let thick mucus be aspirated. Second, a grommet in place.
Dhingra, figs 10.3 and 10.4

**OPERATIVE GUIDES IN THEATRE** 3 operations

**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation**
Tubes: OME with hearing loss or at-risk child, or recurrent AOM
**Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation in the Pediatric Population**
The same operation in children
**Adenoidectomy**
Adenoidectomy: age over 4 with OME, repeat tubes, or nasal or sleep symptoms

### 4.5.3 Adenoidectomy

Added for children over about four, for repeat tube insertion, or when there are nasal or sleep symptoms. It reduces the chance of needing tubes again.

### 4.5.4 Hearing aids

An alternative to surgery, especially in Down syndrome where the canals are narrow and effusion persists.

**GUIDELINES AND NUMBERS (OXFORD TEXTBOOK)**

Effusion peaks at about two years, when around one in five screened children have it, with a second peak at five; up to 90% of children have at least one episode before school. It affects 60 to 85% of children with Down syndrome or cleft palate. Screen-detected effusions clear in about 63% by three months and 88% by a year, but once a **bilateral** effusion has lasted three months, only about a quarter clear by six months and half persist three years later.

**NICE** advises three months of watchful waiting, then ventilation tubes for persistent bilateral effusion with a hearing level in the better ear of 25 to 30 dB HL or worse (averaged over 0.5, 1, 2 and 4 kHz), or with significant developmental, educational or behavioural effects; **hearing aids** are an equal alternative, and are preferred in Down syndrome. NICE does not recommend adenoidectomy with the first set of tubes unless there are persistent or frequent upper respiratory symptoms, whereas the American guideline offers tubes alone under four years and tubes, adenoidectomy or both from four years. Otoscopy by an experienced examiner, pneumatic otoscopy and tympanometry each detect over 90% of effusions; tympanometry is the least specific.

**Tube sequelae:** otorrhoea in about 16% soon after surgery and 26% later; blockage up to 7%; granulation 5%; tympanosclerosis about 32% and focal atrophy 25%; persistent perforation about 2%, rising to about 17% with long-stay tubes. Otorrhoea is best treated with topical antibiotic or antibiotic–steroid drops rather than oral antibiotics, and water precautions make little difference for most children.

**PEARLS**

- Almost every child with a cleft palate has effusion: offer earlier tubes, and keep watching after palate repair.
- Grommet otorrhoea is treated with topical drops, not oral antibiotics.
- Propping an infant up during feeds reduces reflux up the tube.

# 5 CHAPTER FIVE Complications and sequelae

Acute and long term.

## 5.1 Of acute otitis media

- Perforation in about 5%; around 90% heal.
- Acute mastoiditis and subperiosteal abscess.
- Facial palsy and labyrinthitis.
- Meningitis and other intracranial complications.

## 5.2 Of chronic effusion

- Atrophic, retracted drum and atelectasis.
- Necrosis of the long process of the incus.
- Tympanosclerosis.
- Retraction pockets and cholesteatoma.
- Cholesterol granuloma.

## 5.3 Masked disease

Inadequate antibiotic courses can leave a smouldering ear that becomes a persistent effusion or masked mastoiditis.

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 On call

**A CHILD WITH A PAINFUL EAR**

1. Look for complications first: face, balance, mastoid, neck and conscious level.
2. Give analgesia to every child.
3. Perform pneumatic otoscopy; a red drum alone is not enough.
4. Decide antibiotics or observation by age, severity and laterality.
5. Safety-net: return if worse at 48–72 hours, with neck swelling, weakness or drowsiness.

## 6.2 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Stages of acute suppurative otitis media** and the drum at each stage.
- **Why children:** the Eustachian tube and the adenoid.
- **The three bacteria,** and beta-lactamase production.
- **Effusion:** drum appearance, type B tympanogram, 20–40 dB conductive loss.
- **Indications for grommets,** and when to add adenoidectomy.
- **Unilateral effusion in an adult:** examine the nasopharynx.

## 6.3 Common pitfalls

**COMMON PITFALLS**

- Diagnosing otitis media from a red drum in a crying child.
- Withholding analgesia while deciding about antibiotics.

- Treating effusion with decongestants and antihistamines.
- Missing nasopharyngeal carcinoma in an adult with one-sided glue.

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Written for this app from Oxford Textbook of Otolaryngology ch. 84, 88, Oxford Specialist Handbook ch. 16, 23, Oxford Handbook of ENT ch. 6, 17, Dhingra ch. 10 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

## Adenoid Hypertrophy

### IMAGING FINDINGS

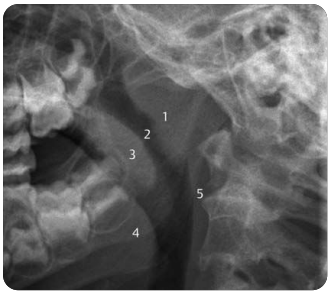
- Symmetric soft tissue in the nasopharyngeal roof and posterior wall
- Striated pattern on T2 with uniform enhancement
- Narrowed nasopharyngeal airway on a lateral film

### TOP DIFFERENTIALS

- Nasopharyngeal carcinoma or lymphoma in an adult
- Thornwaldt cyst
- Juvenile angiofibroma

### CHECKLIST

- In adults, asymmetric or bulky adenoids need endoscopy
- Middle ear effusions




Plain radiograph for adenoid evaluation.

Dunnebier, Imaging for Otolaryngologists, Fig. 7.2

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

# Otomastoiditis and its complications

## IMAGING FINDINGS

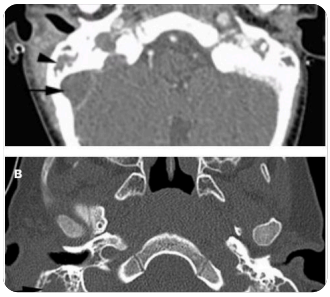
- Opacified middle ear and mastoid
- Coalescence: loss of the mastoid septa and cortical erosion
- Complications: subperiosteal abscess, sigmoid sinus thrombosis, epidural or brain abscess, labyrinthitis

## TOP DIFFERENTIALS

- Uncomplicated effusion (septa intact)
- Cholesteatoma
- Langerhans cell histiocytosis or rhabdomyosarcoma in a child

## CHECKLIST

- Contrast CT or MRI with venography when complications are suspected
- Check the sigmoid sinus, tegmen, labyrinth and petrous apex



Post-contrast CT shows (a) right coalescent mastoiditis (black arrowhead) and ipsilateral sigmoid sinus thrombosis (black arrow) – bot…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.